

TrustLot System Diagrams

1. Entity-Relationship (ER) Diagram

erDiagram

```
APP_USER ||--o{ CAR : "lists"
APP_USER ||--o{ PURCHASE_ORDER : "buys"
APP_USER ||--o{ PURCHASE_ORDER : "sells"
APP_USER ||--o{ SUPPORT_TICKET : "creates"
APP_USER ||--o{ SELLER_REVIEW : "gives"
APP_USER ||--o{ FEEDBACK : "submits"
APP_USER ||--o{ WISHLIST_ITEM : "adds"
APP_USER ||--o{ RECENT_VIEW : "views"

CAR ||--o{ PURCHASE_ORDER : "involves"
CAR ||--o{ WISHLIST_ITEM : "appears-in"
CAR ||--o{ RECENT_VIEW : "appears-in"

PURCHASE_ORDER ||--|| PAYMENT : "has"
PURCHASE_ORDER ||--o{ SUPPORT_TICKET : "relates-to"

SUPPORT_TICKET ||--o{ TICKET_RESPONSE : "contains"

SELLER_REVIEW ||--|| APP_USER : "reviews"

APP_USER {
    int id PK
    string username UK
    string name
    string phoneNumber UK
    string email UK
    string passwordHash
    string role
    timestamp createdAt
    timestamp updatedAt
}

CAR {
    int id PK
    int sellerId FK
    string make
    string model
    int year
    decimal price
    int mileage
    string color
    boolean available
    string approvalStatus
    int viewCount
    string condition
    string fuelType
    string transmission
    string bodyType
    int numberOfOwners
```

```

    int engineCc
    boolean insured
    boolean pucValid
    string description
    timestamp createdAt
}

PURCHASE_ORDER {
    int id PK
    int buyerId FK
    int sellerId FK
    int carId FK
    int paymentId FK
    string status
    boolean fraudAlert
    timestamp createdAt
}

PAYMENT {
    string id PK
    string status
    decimal amount
    string method
    string gatewayName
    string gatewayTransactionId
    string failureReason
    timestamp paidAt
}

SUPPORT_TICKET {
    int id PK
    int userId FK
    string subject
    string description
    string status
    timestamp createdAt
    timestamp updatedAt
}

TICKET_RESPONSE {
    int id PK
    int ticketId FK
    int senderId FK
    string message
    timestamp respondedAt
}

FEEDBACK {
    int id PK
    int userId FK
    string message
    timestamp createdAt
}

```

```

SELLER_REVIEW {
    int id PK
    int reviewerId FK
    int sellerId FK
    int rating
    string comment
    timestamp createdAt
}

WISHLIST_ITEM {
    int id PK
    int buyerId FK
    int carId FK
    timestamp addedAt
}

RECENT_VIEW {
    int id PK
    int userId FK
    int carId FK
    timestamp viewedAt
}

```

2. Data Flow Diagram (DFD) - Level 0

```

graph TB
    User[" User<br/>(Buyer/Seller)"]
    Admin[" Admin<br/>(Moderator)"]
    System[" TrustLot<br/>System"]
    PaymentGW[" Payment<br/>Gateway"]
    EmailService[" Email<br/>Service"]
    Database[" Database<br/>(MySQL/Derby)"]

    User -->|Browse, List, Purchase| System
    Admin -->|Approve, Reject, Manage| System
    System -->|Process Payment| PaymentGW
    PaymentGW -->|Payment Status| System
    System -->|Send Notifications| EmailService
    System -->|Read/Write Data| Database
    Database -->|Data| System
    System -->|Car Listings, Orders, Reviews| User
    System -->|Dashboard, Approvals| Admin
    EmailService -->|Emails| User

```

3. Data Flow Diagram (DFD) - Level 1: User & Car Management

```

graph LR
    subgraph "1. User Layer"
        Buyer[" Buyer<br/>Users"]
        Seller[" Seller<br/>Users"]
    end
    end

```

```

subgraph "2. Authentication & Session"
    Auth[" Auth<br/>Service"]
    SessionMgmt[" Session<br/>Management"]
end

subgraph "3. Car Management"
    CarService[" Car<br/>Service"]
    CarRepo[" Car<br/>Repository"]
end

subgraph "4. Data Storage"
    Database[( Database)]
end

Buyer -->|Register/Login| Auth
Seller -->|Register/Login| Auth
Auth -->|Create Session| SessionMgmt
SessionMgmt -->|Token| Buyer
SessionMgmt -->|Token| Seller

Seller -->|List Car| CarService
Buyer -->|Browse Cars| CarService
CarService -->|Query/Store| CarRepo
CarRepo -->|CRUD Operations| Database
Database -->|Car Data| CarRepo
CarRepo -->|Car Info| CarService
CarService -->|Available Cars| Buyer

```

4. Data Flow Diagram (DFD) - Level 1: Purchase & Payment

```

graph LR
    subgraph "1. Purchase Initiation"
        Buyer[" Buyer"]
    end

    subgraph "2. Purchase Service"
        PurchaseService[" Purchase<br/>Service"]
        PaymentService[" Payment<br/>Service"]
    end

    subgraph "3. External Services"
        PaymentGW[" Payment<br/>Gateway"]
    end

    subgraph "4. Admin Approval"
        AdminService[" Admin<br/>Service"]
        Admin[" Admin"]
    end

    subgraph "5. Seller Approval"
        SellerService[" Seller<br/>Approval"]
        Seller[" Seller"]
    end

```

```

end

subgraph "6. Order Management"
    OrderRepo[" Order<br/>Repository"]
end

subgraph "7. Storage"
    Database[( Database)]
end

Buyer -->|Initiate Purchase| PurchaseService
PurchaseService -->|Process Payment| PaymentService
PaymentService -->|Send Payment| PaymentGW
PaymentGW -->|Payment Status| PaymentService
PaymentService -->|Store Order| OrderRepo
OrderRepo -->|PENDING_ADMIN_APPROVAL| Database

Admin -->|Review Orders| AdminService
AdminService -->|Query Pending| OrderRepo
OrderRepo -->|Order Details| AdminService
AdminService -->|Update Status| OrderRepo
OrderRepo -->|PENDING_SELLER_APPROVAL| Database

Seller -->|Review Sales| SellerService
SellerService -->|Get Pending| OrderRepo
OrderRepo -->|Sale Details| SellerService
SellerService -->|Approve/Reject| OrderRepo
OrderRepo -->|Update Status| Database

```

5. Data Flow Diagram (DFD) - Level 1: Support & Reviews

```

graph LR
    subgraph "1. User Actions"
        Buyer[" Buyer"]
        Seller[" Seller"]
    end

    subgraph "2. Support System"
        TicketService[" Support<br/>Ticket Service"]
        TicketRepo[" Ticket<br/>Repository"]
    end

    subgraph "3. Review System"
        ReviewService[" Review<br/>Service"]
        ReviewRepo[" Review<br/>Repository"]
    end

    subgraph "4. Admin Response"
        Admin[" Admin"]
        AdminService[" Admin<br/>Message Service"]
    end

    subgraph "5. Storage"

```

```

    Database[( Database)]
end

subgraph "6. Notifications"
    EmailService[" Email<br/>Service"]
end

Buyer -->|Create Ticket| TicketService
Buyer -->|Add Message| TicketService
TicketService -->|Store/Query| TicketRepo
TicketRepo -->|Read/Write| Database

Admin -->|View Tickets| AdminService
AdminService -->|Query Tickets| TicketRepo
TicketRepo -->|Ticket Data| AdminService
AdminService -->|Add Response| TicketRepo
TicketRepo -->|Update| Database
TicketService -->|Send Email| EmailService

Buyer -->|Submit Review| ReviewService
Seller -->|View Reviews| ReviewService
ReviewService -->|Store/Query| ReviewRepo
ReviewRepo -->|Read/Write| Database
ReviewService -->|Notify| EmailService
EmailService -->|Email Notification| Buyer

```

6. Data Flow Diagram (DFD) - Level 1: Admin Dashboard

```

graph LR
    subgraph "1. Admin Interface"
        Admin[" Admin<br/>User"]
    end

    subgraph "2. Admin Services"
        DashboardService[" Dashboard<br/>Service"]
        ApprovalService[" Approval<br/>Service"]
        UserService[" User<br/>Management"]
    end

    subgraph "3. Data Aggregation"
        Analytics[" Analytics<br/>Engine"]
        Stats[" Statistics<br/>Calculator"]
    end

    subgraph "4. Data Access"
        CarRepo[" Car<br/>Repository"]
        OrderRepo[" Order<br/>Repository"]
        UserRepo[" User<br/>Repository"]
    end

    subgraph "5. Storage"
        Database[( Database)]
    end

```

```

Admin -->|View Dashboard| DashboardService
DashboardService -->|Query Stats| Analytics
Analytics -->|Query Data| CarRepo
Analytics -->|Query Data| OrderRepo
Analytics -->|Query Data| UserRepo

CarRepo -->|Car Data| Analytics
OrderRepo -->|Order Data| Analytics
UserRepo -->|User Data| Analytics

Analytics -->|Calculate| Stats
Stats -->|Dashboard Data| DashboardService
DashboardService -->|Charts & Metrics| Admin

Admin -->|Approve Cars| ApprovalService
ApprovalService -->|Update Car| CarRepo
CarRepo -->|Persist| Database

Admin -->|Edit Users| UserService
UserService -->|Update User| UserRepo
UserRepo -->|Persist| Database

```

7. Complete System Data Flow

graph TB

```

User[" Users<br/>(Buyers/Sellers)"]
Admin[" Admin"]

```

```

subgraph "API Layer"
    AuthAPI[" Auth API"]
    CarAPI[" Car API"]
    PurchaseAPI[" Purchase API"]
    AdminAPI[" Admin API"]
    TicketAPI[" Support API"]
    ReviewAPI[" Review API"]
end

```

```

subgraph "Business Logic Layer"
    AuthService["Auth Service"]
    CarService["Car Service"]
    PurchaseService["Purchase Service"]
    PaymentService["Payment Service"]
    ApprovalService["Approval Service"]
    TicketService["Ticket Service"]
    ReviewService["Review Service"]
end

```

```

subgraph "Data Access Layer"
    UserRepo["User Repo"]
    CarRepo["Car Repo"]
    OrderRepo["Order Repo"]
    PaymentRepo["Payment Repo"]

```

```

    TicketRepo["Ticket Repo"]
    ReviewRepo["Review Repo"]
end

subgraph "External Services"
    PaymentGW[" Payment Gateway"]
    EmailService[" Email Service"]
end

subgraph "Database"
    DB[( Database<br/>MySQL/Derby)]
end

User -->|API Calls| AuthAPI
User -->|API Calls| CarAPI
User -->|API Calls| PurchaseAPI
User -->|API Calls| TicketAPI
User -->|API Calls| ReviewAPI

Admin -->|API Calls| AdminAPI
Admin -->|API Calls| CarAPI

AuthAPI -->|Authenticate| AuthService
CarAPI -->|Business Logic| CarService
PurchaseAPI -->|Business Logic| PurchaseService
AdminAPI -->|Business Logic| ApprovalService
TicketAPI -->|Business Logic| TicketService
ReviewAPI -->|Business Logic| ReviewService

PurchaseService -->|Process Payment| PaymentService
PaymentService -->|External Call| PaymentGW
PaymentGW -->|Payment Status| PaymentService

AuthService -->|CRUD| UserRepo
CarService -->|CRUD| CarRepo
PurchaseService -->|CRUD| OrderRepo
PaymentService -->|CRUD| PaymentRepo
TicketService -->|CRUD| TicketRepo
ReviewService -->|CRUD| ReviewRepo
ApprovalService -->|CRUD| CarRepo
ApprovalService -->|CRUD| OrderRepo

UserRepo -->|Persist| DB
CarRepo -->|Persist| DB
OrderRepo -->|Persist| DB
PaymentRepo -->|Persist| DB
TicketRepo -->|Persist| DB
ReviewRepo -->|Persist| DB

TicketService -->|Notifications| EmailService
PurchaseService -->|Notifications| EmailService
EmailService -->|Send Emails| User

```

Diagram Descriptions

ER Diagram

- **Entities:** 10 main entities (AppUser, Car, PurchaseOrder, Payment, etc.)
- **Relationships:** One-to-many (user lists cars), Many-to-one (purchase involves car), One-to-one (order has payment)
- **Keys:** Primary keys (id), Foreign keys (FK), Unique keys (UK) marked
- **Attributes:** All important fields including timestamps and status fields

DFD Level 0

- **Context Diagram:** Shows system boundary with external actors
- **External Entities:** Users, Admin, Payment Gateway, Email Service
- **System:** Central TrustLot system processing all transactions

DFD Level 1 - User & Car Management

- **Authentication Flow:** User registration → Auth Service → Session Management
- **Car Management:** Listing cars, browsing cars → Car Service → Database

DFD Level 1 - Purchase & Payment

- **Purchase Workflow:** Buyer initiates → Payment processing → Admin approval → Seller approval
- **State Transitions:** PENDING_ADMIN_APPROVAL → PENDING_SELLER_APPROVAL → APPROVED

DFD Level 1 - Support & Reviews

- **Support Workflow:** Create ticket → Admin response → Email notification
- **Review System:** Submit review → Store → Email notification

DFD Level 1 - Admin Dashboard

- **Data Aggregation:** Queries from multiple repositories
- **Analytics:** Statistics and charts generation
- **Admin Actions:** Approvals and user management

Complete System DFD

- **Layered Architecture:** API → Business Logic → Data Access → Database
- **All Flows:** Shows complete data flow through entire system
- **External Integration:** Payment Gateway and Email Service integration